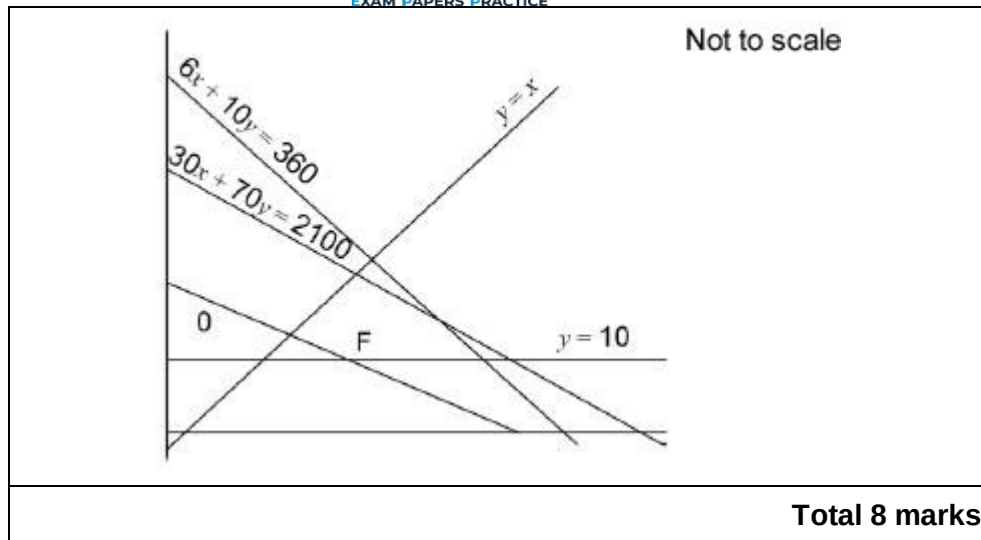


Mark schemes

Q1.

Marking Instructions	AO	Marks	Typical Solution
Introduces two variables and defines at least one of them as 'number of'	AO3.3	B1	x = number of pine tables y = number of oak tables
Finds correctly an inequality for the problem by considering the costs (PI)	AO3.1b	B1	$30x + 70y \leq 2100$ OE
Finds correctly an inequality for the problem by considering the labour (PI)	AO1.1b	B1	$6x + 10y \leq 360$ OE
Finds correctly three inequalities for the problem by considering the wholesaler (PI)	AO1.1b	B1	$y \leq x, y \geq 10, x \geq 0$
Correctly plots the line $6x + 10y = 360$ OE through (0, 36) and (60, 0) AND the line $30x + 70y = 2100$ OE through (0, 30) and (70, 0)	AO1.1b	B1	*See diagram below
Correctly and accurately plots 'their' $y = x$ AND 'their' $y = 10$	AO1.1b	B1	*See diagram below
Identifies and labels 'their' feasible region	AO2.2a	B1	*See diagram below
Uses the optimal vertex of the feasible region and clearly states the solution to the problem in the context of the question	AO3.2a	A1 CAO	Uses an objective line with a gradient of $-8/15$ 35 pine tables and 15 oak tables (Profit = £2525)



Q2.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Formulates correctly one non-trivial inequality	AO3.1b	B1	$3x + 2y + z \leq 360$
	Formulates correctly a second non-trivial inequality	AO3.1b	B1	$40x + 20y + 5z \leq 2500$
	States correctly all three trivial inequalities	AO1.2	B1	$x \geq 0, y \geq 0, z \geq 0$

(b) (i)	Translates 'their' inequalities into a simplex tableau with two slack variables	AO3.1a	M1	See diagram on the next page
	Identifies a correct pivot from 'their' tableau	AO1.1b	A1	
	Uses the simplex algorithm correctly to modify the 'their' two non-pivot rows	AO1.1a	M1	
	Identifies the correct pivot in 'their' modified tableau	AO1.1b	A1F	
	Uses the simplex algorithm correctly to modify 'their' two non-pivot rows	AO1.1a	M1	
	Makes a correct interpretation of 'their' final tableau in the context of the problem, provided the	AO3.2a	E1	To maximise the profit, the company should repair and sell 28 monitors, 0 hard drives and 276

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	objective row is non-negative			keyboards each month																																																																						
(ii)	States no further use of the simplex algorithm is required and explains why	AO2.4	R1	The objective row of the final tableau being non-negative numbers.																																																																						
<table border="1"> <thead> <tr> <th><i>P</i></th> <th><i>x</i></th> <th><i>y</i></th> <th><i>z</i></th> <th><i>s</i></th> <th><i>t</i></th> <th>value</th> </tr> </thead> <tbody> <tr> <td>1</td> <td>-80</td> <td>-35</td> <td>-15</td> <td>0</td> <td>0</td> <td>0</td> </tr> <tr> <td>0</td> <td>3</td> <td>2</td> <td>1</td> <td>1</td> <td>0</td> <td>360</td> </tr> <tr> <td>0</td> <td>40</td> <td>20</td> <td>5</td> <td>0</td> <td>1</td> <td>2500</td> </tr> <tr> <td>1</td> <td>0</td> <td>5</td> <td>-5</td> <td>0</td> <td>2</td> <td>5000</td> </tr> <tr> <td>0</td> <td>0</td> <td>0.5</td> <td>0.625</td> <td>1</td> <td>-0.075</td> <td>172.5</td> </tr> <tr> <td>0</td> <td>1</td> <td>0.5</td> <td>0.125</td> <td>0</td> <td>0.025</td> <td>62.5</td> </tr> <tr> <td>1</td> <td>0</td> <td>9</td> <td>0</td> <td>8</td> <td>1.4</td> <td>6380</td> </tr> <tr> <td>0</td> <td>0</td> <td>0.8</td> <td>1</td> <td>1.6</td> <td>-0.12</td> <td>276</td> </tr> <tr> <td>0</td> <td>1</td> <td>0.4</td> <td>0</td> <td>-0.2</td> <td>0.04</td> <td>28</td> </tr> </tbody> </table>					<i>P</i>	<i>x</i>	<i>y</i>	<i>z</i>	<i>s</i>	<i>t</i>	value	1	-80	-35	-15	0	0	0	0	3	2	1	1	0	360	0	40	20	5	0	1	2500	1	0	5	-5	0	2	5000	0	0	0.5	0.625	1	-0.075	172.5	0	1	0.5	0.125	0	0.025	62.5	1	0	9	0	8	1.4	6380	0	0	0.8	1	1.6	-0.12	276	0	1	0.4	0	-0.2	0.04	28
<i>P</i>	<i>x</i>	<i>y</i>	<i>z</i>	<i>s</i>	<i>t</i>	value																																																																				
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(iii)	Introduces a new inequality for the hard drive hardware, ensuring that at least some hard drives are required to be repaired and sold	AO3.4	E1	As $y = 0$, enforce some hard drives to be repaired by requiring that, for instance, $y \geq 10$																																																																						
Total 11 marks																																																																										

Q3.

$$2x + 3y + 5z \leq 400$$

$$3x + 4y + 3z \leq 400$$

Both

B1

$$(6x + 2y + 2z \leq 400)$$

$$\Rightarrow 3x + y + z \leq 200$$

B1

$$11x + 9y + 10z \geq 1000$$

B1

$$\text{their}(2x + 3y + 5z) > \text{their}(3x + 4y + 3z)$$

$$\text{Condone} \geq$$

M1

$$2z > x + y$$

oe

A1

CAO

$$6x + 2y + 2z \leq \frac{4}{10} (11x + 9y + 10z)$$

Condone <

Allow numerical values to $\frac{4}{10}$

M1

$$16x + 16y + 20z \leq 0 \text{ oe}$$

A1

$$4x \leq 4y + 5z$$

A1

CAO

[8]

Q4.

(a) $6x + 4y + 3z \geq 420$

B1

$$6x + 6y + 4z \geq 480$$

oe

B1

$$6x + 4y + 4z \leq 720$$

oe

B1

3

(b) (i) $(y = z)$

$$6x + 4y + 3z \geq 420 \Rightarrow 6x + 7y \geq 420$$

Must see this substitution

B1

$$6x + 10y \geq 480 \Rightarrow 3x + 5y \geq 240 \quad \text{oe}$$

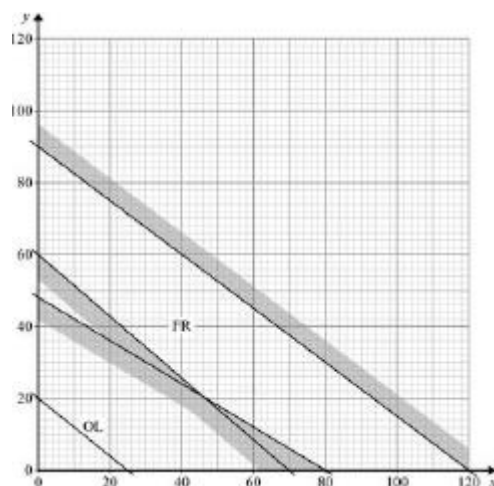
$$6x + 8y \geq 720 \Rightarrow 3x + 4y \geq 360 \quad \text{oe}$$

Both other inequalities correct, condone direct substitution into simplified versions of part (a)

B1

2

(ii)



Accuracy: All lines must be ruled, correct to within $\frac{1}{2}$ square
BOTH horizontally and vertically
Correct at (0, 60) and (70, 0)

Correct at (0, 48) and (80, 0)

Correct at (0, 90) and (120, 0)

FR labelled, MUST have scored previous 3 marks

Condone omission of shading on axes
OL, drawn, with gradient -0.8 or -1.25

Gradient -0.8

B1

B1

B1

B1

M1

A1

6

(iii) (Max profit =) £480
Including '£'

B1

120 gold, 0 silver, 0 bronze
All 3 must be stated

B1

2

(c) (Max profit =) £1080
Including '£'

B1

0 gold, 90 silver, 90 bronze

If B0 scored then B1 for $x = 0$ and $y = 90$, PI

B2

3

[16]

Q5.

(a)

$$\left. \begin{array}{l} x \geq 100, y \geq 200 \\ x + y + z \geq 400 \end{array} \right\} \text{OE}$$

B1

$$4x + 3y + 4z \leq 1800 \quad \text{OE}$$

B1

$$y \geq \frac{40}{100} (x + y + z) \quad \text{OE}$$

B1

3

(b) (i) $(x = 2z)$

$$\left. \begin{array}{l} x + y + \frac{1}{2}x \geq 400 \\ \Rightarrow 3x + 2y \geq 800 \end{array} \right\}$$

Correct substitution and fully simplifying 1 inequality
(must see evidence: either replacing z or multiplying inequality)

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M1

$$\left. \begin{array}{l} 4x + 3y + 2x \leq 1800 \\ 6x + 3y \leq 1800 \\ 2x + y \leq 600 \end{array} \right\}$$

As above 'in 2nd inequality'

A1

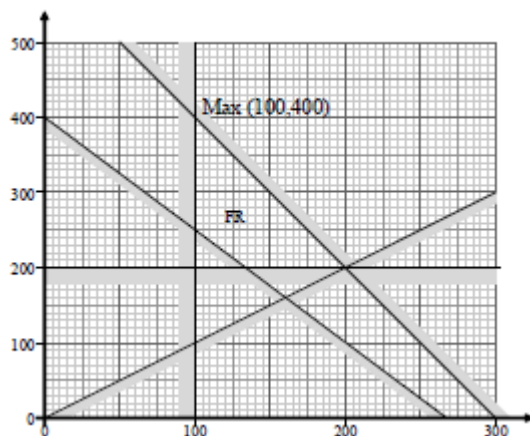
$$\left. \begin{array}{ll} 5y \geq 2x + 2y + x & 5y \geq 2x + 2y + x \\ 3y \geq 3x & 3y \geq 3x \\ y \geq x & y \geq x \end{array} \right\}$$

As above 'in 3rd inequality'

A1

3

(ii)



Each line must be straight to have the B mark available.
For all lines, must be correct to half square horizontal and vertical at the indicated vertices.

$$x = 100, y = 200$$

B1

$y = x$ line
through (100, 100) and (200, 200)

B1

$2x + y = 600$ line
through (100, 400) and (200, 200)

B1

$3x + 2y = 800$ line
through (100, 250) and (200, 100)

B1

Feasible Region, all lines correct and region labelled
(condone no shading, ignore 'poor' shading)

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B1

5

(iii) (Max) $y + \frac{3}{2}x$
PI by objective line with gradient -1.5

M1

$$(\quad = 400 + 150) = 550$$

A1

2

(iv) Buys 100 soft
400 medium
50 firm

B1

Q6.

(a) $6x + 9y + 9z \leq 600$

*Any of the three inequalities correct
(un)simplified, condone strict inequalities*

M1

$$2x + 3y + 3z \leq 200$$

CAO

A1

$$9x + 6y + 9z \leq 600$$

$$3x + 2y + 3z \leq 200$$

CAO

A1

$$6x + 12y + 18z \geq 480$$

$$x + 2y + 3z \geq 80$$

CAO

A1

4

(b) (i) $(z = y)$

$$2x + 3y + 3y \leq 200 \text{ or } 2x + 6y \leq 200$$

*Correctly substitute into **this** inequality-
either simplified or unsimplified form*

M1

$$x + 3y \leq 100$$

AG

$$3x + 2y + 3z \leq 200$$

*Correctly substitute into **this** inequality-
either simplified or unsimplified form*

$$(\Rightarrow) 3x + 5y \leq 200$$

AG

$$x + 2y + 3z \geq 80$$

*Correctly substitute into **this** inequality-
either simplified or unsimplified form*

$$(\Rightarrow) x + 5y \geq 80$$

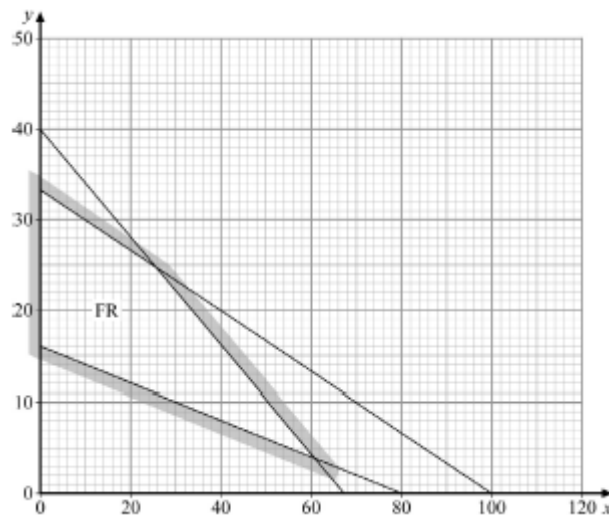
AG

*All correct – must link their original inequality
to the stated answers*

A1

2

- (ii) Each line must be straight to have the B mark available.
For all lines, must be correct to $\frac{1}{2}$ square horizontal and vertical at the indicated vertices.



Line through (10, 30) and (40, 20)

B1

Line through (50, 10) and (0, 40)

B1

Line through (80, 0) and (0, 16)

B1

FR, must have all lines correct and labelled
region (condone no shading)

B1

4

- (iii) Max $x + 2y$ PI

If no statement (PI), then check OL on diagram,
which must be correct for M1

M1

$$\text{Max } (= 25 + 50) = 75$$

Note: 75 with no working 2/2

A1

2

- (iv) 25 basic, 25 standard, 25 luxury

Condone "25 of each type" ONLY if (b)(iii)
fully correct

Note $x = 25 = y = z$ B0

B1F

1

[13]

Q7.

(a) $x + 5y \geq 25$ OE
ISW

B1

$2x + 15y \geq 60$ OE
ISW

B1

$x + 25y \geq 40$ OE
ISW

B1

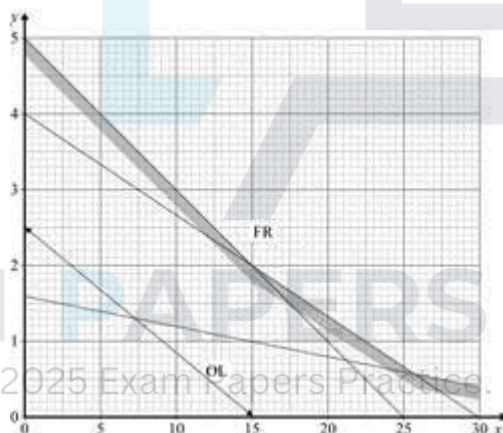
(C =) $2.5x + 15y$

ISW; condone $250x + 1500y$, but not any other multiples

B1

4

(b) (i)



Note: all points need to be correct to within half a square horizontally and vertically
Line through (0, 5) and (25, 0)

B1

Line through (0, 4) and (30, 0)

B1

Line through (15, 1) and (30, 0.4)

B1

FR, must have all lines correct and labelled region (condone no shading)

B1

Objective line drawn, gradient of $-\frac{1}{6}$ or -6

M1

$$\text{Gradient} = -\frac{1}{6}$$

A1

6

(ii) 15 DIY, 2 trade

B1

1

(iii) (Cost) £67.50

Condone 6750p, £67.5

B1

1

[12]

Q8.

$$2x + 3y + 4z \leq 360$$

$$3x + y + 5z \leq 300$$

$$4x + 3y + 2z \leq 400$$

B2,1,0

$$2x + 3y + 4z (>) 3x + y + 5z$$

Their A (>) their B

M1

$$2y > x + z$$

OE

A1

$$5x + 4y + 9z (\geq) 4x + 3y + 2z$$

Their A + B ($\geq$) their C

M1

$$x + y + 7z \geq 0$$

OE

A1

$$4x + 3y + 2z (\geq) \frac{40}{100} (9x + 7y + 11z)$$

Their C ($\geq$) 40% of their total OE

M1

$$2x + y \geq 12z$$

OE

A1

8

[8]

Q9.

(a) $x \geq 190, \quad y \geq 50, \quad z \geq 50$ oe

B1

$x + y + z \geq 300$ oe

B1

$2.5x + 2y + 2z \leq 1000$ oe

B1

$(5x + 4y + 4z \leq 2000$

$x \geq \frac{60}{100}(x + y + z)$

oe

B1

Strict inequalities: penalise first two instances only

$(2x \geq 3y + 3z)$

4

(b) (i) $y = z$

$x \geq 190, \quad y \geq 50$

$x + 2y \geq 300$

oe

$x + y = y \geq 300$ or $5x + 4y + 4y \leq 2000$

or $2x \geq 3y + 3y$

ie at least one clear line of working

showing substitution of $y = z$

M1

$5x + 8y \leq 2000$

$2x \geq 6y$

$\left(y \leq \frac{1}{3}x\right)$

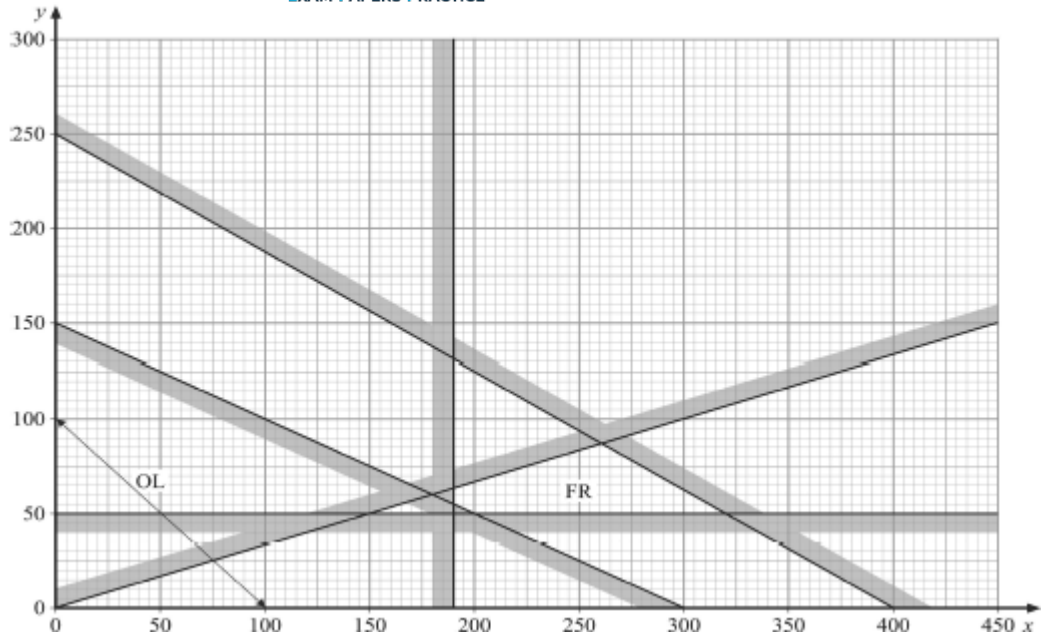
oe

AG All correct (3 'or' become 'and')

A1

2

(ii)



For all lines must be correct to $\frac{1}{2}$ square
horizontal or vertical

$$x = 190, y = 50$$

B1

through (0,150) and (300,0)

B1

through (0,250) and (400,0)

B1

$$y = mx \text{ through } (0,0)$$

M1

through (300,100)

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A1

Region must have all lines correct and
labelled region (condone lack of shading)

B1

A correct objective line

B1

7

(iii)
$$P = \frac{1}{2}x + \frac{1}{4}y + \frac{1}{4}z \quad \text{or} \quad \frac{1}{2}x + \frac{1}{2}y$$

PI

M1

Max at (320, 50)

B1

Profit $(160 + 25) = £185$

A1

Buys 320 slow, 50 medium, 50 fast

B1

Note: (with no working) £185 3/4
320 slow, 50 medium, 50 fast 2/4
320 slow, 50 medium, 50 fast and £185 4/4

4

[17]

Q10.

(a) $x + y + z \geq 110$

–1 for strict inequalities (max)

–1 for using g, p, s instead of x, y, z

(max)

$y \geq x$

B1

$y + z \leq 150$

B1

$16x + 8y + 24z \leq 3120$ ISW

$(2x + y + 3z \leq 390)$

B1

B1

$(P \Rightarrow) 70x + 30y + 50z$

B1

5

(b) (i) $z = 30$

Justify by correctly substituting into at least one of their inequalities

M1

$x + y \geq 80$ (or $x + y + 30 \geq 110$)

$(y \geq x)$

$y \leq 120$ (or $y + 30 \leq 150$)

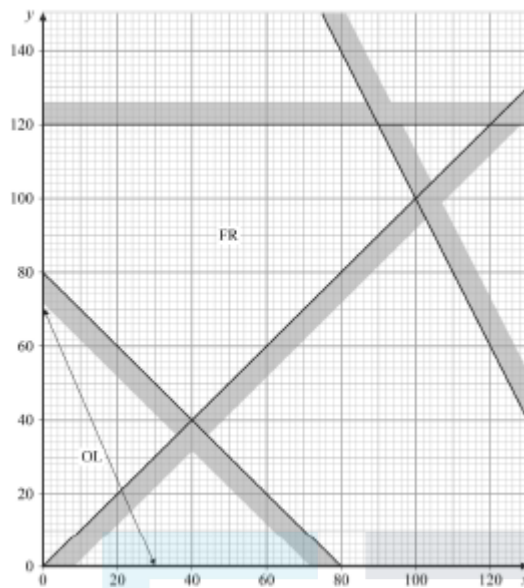
$2x + y \leq 300$ (or $2x + y + 90 \leq 390$ OE)

Correctly substituting into all 3 inequalities

A1

$(P = 70x + 30y + 1500)$

(ii)



$$y = 120$$

B1

$$x + y = 80$$

B1

$$y = x \text{ correct at } (40, 40) \text{ and } (100, 100)$$

B1

$$2x + y = 300, \text{ -ve gradient with one correct point in the interval } 80 \leq x \leq 120$$

M1

$$\text{Correct at } (100, 100) \text{ and } (90, 120)$$

A1

$$\text{Correct region labelled}$$

B1

$$\text{OL: gradient of } -\frac{7}{3} \text{ or } -\frac{3}{7}$$

M1

$$\text{Gradient} = -\frac{7}{3}$$

A1

(iii) Considering (90, 120) and/or (100, 100)
Ignore other points being considered

M1

(£) 11500

A1

100 goats, 100 pigs, 30 sheep
CAO

A1

3

[18]

Q11.

Working must be in x , y and z
Equalities can only score M marks
Strict inequalities: -1 first error only

(a) $6x + 4y + 2z \leq 240$

M1

$3x + 2y + z \leq 120$

CAO

A1

$6x + 3y + 9z \leq 300$

M1

$2x + y + 3z \leq 100$

CAO

A1

$12x + 18y + 6z \leq 900$

M1

$2x + 3y + z \leq 150$

CAO

A1

$12x + 18y + 6z \geq 2(6x + 3y + 9z)$

OE

M1

$y \geq z$

CSO; OE in simplified form eg $y - z \geq 0$

A1

8

- (b) (i) $(z = x)$
 $4x + 2y \leq 120$ OE or $3x + 3y \leq 150$ OE
 $4x + 2y \leq 120$ OE $\Rightarrow 2x + y \leq 60$ **AG**
 $3x + 3y \leq 150$ OE $\Rightarrow x + y \leq 50$ **AG**
 $5x + y \leq 100, y \geq x$ **AG**

*Correct unsimplified subst $x = z$ into
 either of these 2 correct inequs. (seen)*

M1

Both correct and simplified

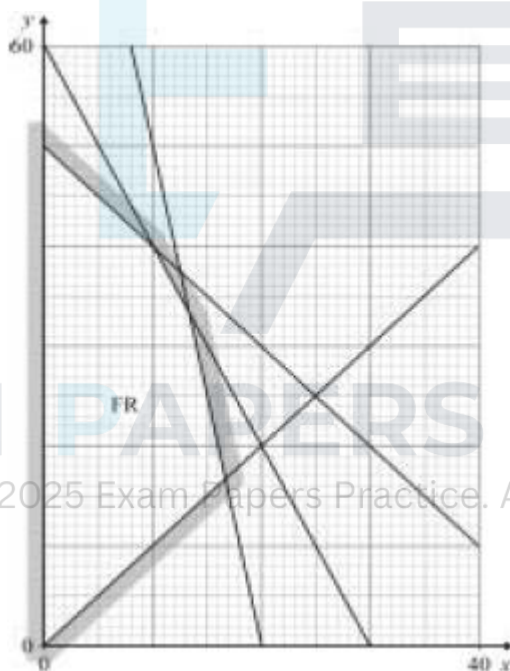
A1

Correct subst $x = z$ into 4 correct inequs.

A1

3

(ii)



Line 1 correct at (0, 50) (25, 25)

B1

Line 2 correct at (10, 50) (20, 0)

B1

Line 3 correct at (0, 60) (30, 0)

B1

Line 4 correct at (0, 0) (25, 25)

B1

Each line correct to $\frac{1}{2}$ square,

horizontally or vertically
FR, must have all lines correct **and**
labelled region (condone no shading)

B1
5

(iii) $N = x + y + z = 2x + y$
Stated or PI

M1

Max = 60
CSO; **SC** unsupported 60 scores 2/2

A1
2

(iv) 10, 40, 10
11, 38, 11
12, 36, 12
13, 34, 13

Any correct; may be earned in part (iii)

B1

3 correct

B1

4 correct **and** no extras

B1
3

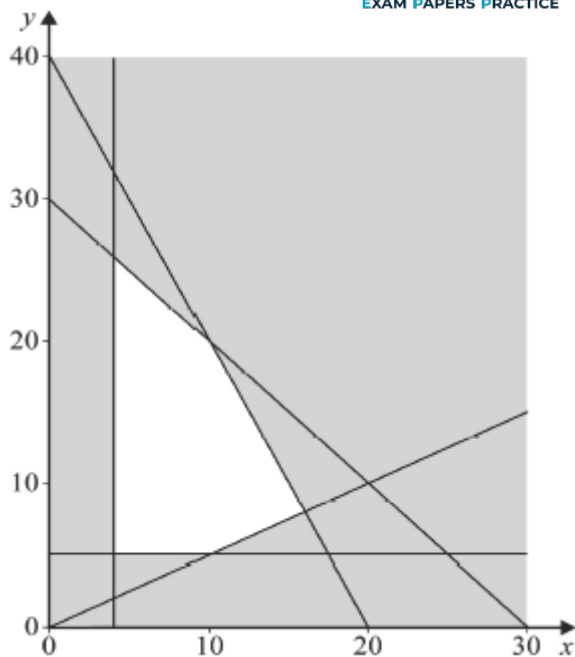
[21]

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Q12.

(a)



$$y = 5, x = 4$$

B1

$$x + y = 30$$

B1

$$2x + y = 40$$

B1

$$y = \frac{1}{2}x$$

B1

feasible region CAO

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B1

5

(b) (i) Max at $(16, 8) = 56$

Extreme point within $\frac{1}{2}$ square of their region

M1A1

2

(ii) Max at $(4, 26) = 82$

Extreme point within $\frac{1}{2}$ square of their region

M1A1

2

[9]

Q13.

$$\left. \begin{array}{l} 2x + 4y + 3z \leq 360 \\ 3x + 2y + 4z \leq 270 \\ x + 3y + 5z \leq 450 \end{array} \right\}$$

Any correct LHS in inequality

M1

OE

A2,1,0

$$6x + 9y + 12z \geq 720$$

M1

$$\Rightarrow x + 3y + 4z \geq 240$$

Allow further correct simplification

A1

$$2x + 4y + 3z \geq \frac{2}{5}(6x + 9y + 12z)$$

Must have 3 parts correct

M1A1

$$2y \geq 2x + 9z \quad \text{OE}$$

Allow further correct simplification

A1

EXAM PAPERS PRACTICE 8 [8]

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Q14.

(a) **All inequalities must be as below**

$$x \leq 100, y \leq 80$$

Both

B1

$$x + y \geq 60$$

B1

$$x < y$$

B1

$$2x + 8y \geq 320$$

OE

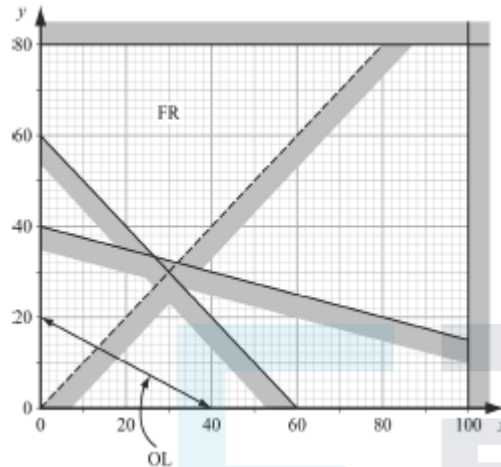
B1

(minimise $C \Rightarrow 1.5x + 3y$)

B1

5

(b)



$x = 100, y = 80$
within $\frac{1}{2}$ square
from $(0, 0)$ to $(80, 80)$

Other lines
within $\frac{1}{2}$ square
from $(0, 0)$ to $(80, 80)$

B1

B1 $\times$ 3

Feasible Region CAO (must have scored
B4 for drawing lines)
(condone $x = y$ as solid line)

B1

An Objective Line with gradient -0.5

B1

6

(c) Considering an extreme point in their region

M1

Min at intersect of $x + y = 60$

$$x + 4y = 160$$

PI by indication on diagram or

$$x = 26\frac{2}{3} \quad y = 33\frac{1}{3}$$

A1

Considering a pair of integer values where
 $26 \leq x \leq 28$, $32 \leq y \leq 34$

M1

(C =) £141 at (26, 34)
 or £141 at (28, 33)

A1

4

[15]

Q15.

- (a) $1000x + 500y \leq 9000$
 $(2x + y \leq 18)$

B1

1

- (b) $x \geq 2$, $y \geq 5$

B1

$y \geq 2x$

B1

$y \leq 3x$

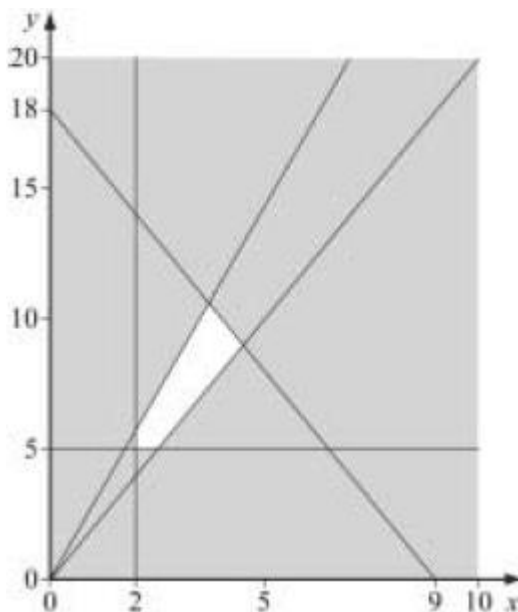
B1

-1 for strict inequalities
 -1 for 'w's and 'l's

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3

- (c)



$$x = 2, y = 5$$

B1

$$2x + y = 18$$

B1

$$\text{Line } y = mx$$

M1

$$y = 2x$$

A1

$$y = 3x$$

A1

Feasible region

B1

6

- (d) Considering an extreme point on their f.r.

Extreme point - vertex

M1

$$x = 4.5$$

A1

$$y = 9$$

A1

3

[13]

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Q16.

- (a) $5x + 10y \leq 1500$ (balloons)
 $\Rightarrow x + 2y \leq 300$

E1

$$32x + 8y \leq 4000$$
 (sweets)

E1

$$\Rightarrow 4x + y \leq 500$$

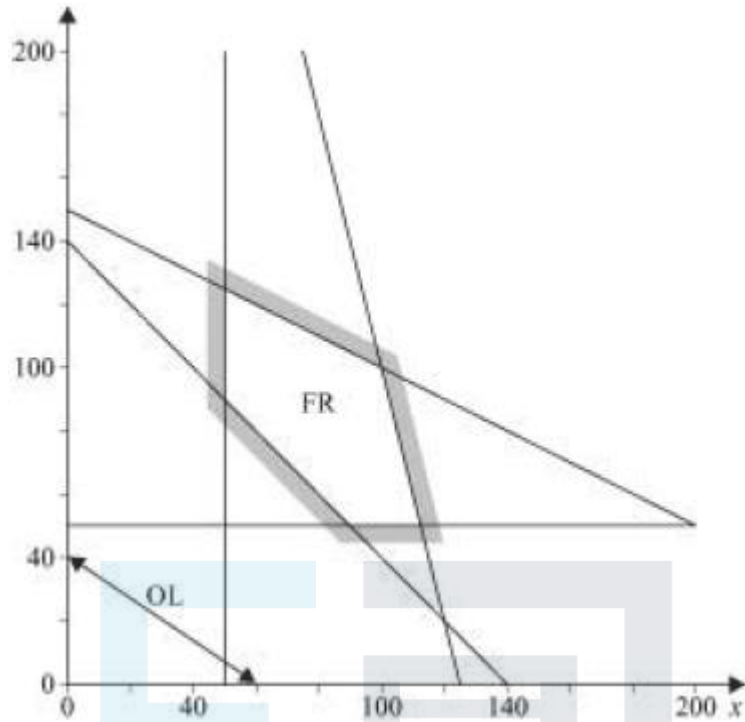
$x \geq 50, y \geq 50$, at least 50 of each

E1

$$x + y \geq 140$$
, at least 140 in total

E1

(b) (i)



$x = 50, y = 50$

$x + y = 140$

B1

B1

Negative gradient (either)

M1

$4x + y = 500$

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A1

$x + 2y = 300$

A1

Feasible region

B1

Objective line drawn

M1A1

8

(ii) Maximum (100,100)

Considering extreme point on their region

M1

= £200

A1

2

(iii) Minimum (90,50)

Considering extreme minimum point on their region

M1

$$= £132$$

A1

2

[16]

Q17.

$$5x + 4j + 3z < 180 \quad \text{oe}$$

B1

$$6x + 4j + 5z < 120 \quad \text{oe}$$

B1

$$4x + 2y + 3z < 90 \quad \text{oe}$$

B1

$$x > y$$

B1

$$y > z$$

B1

EXAM PAPERS PRACTICE

$$x > \frac{40}{100} (x + y + z)$$

x ≥ 40% (their total)

M1

A1

$$3x \geq 2y + 2z$$

A1

8

[8]

Q18.

(a) $10 \leq x \leq 80$

Strict inequalities –1 (or using p, c)

B1

$$5 \leq y \leq 40$$

B1

$$x + y \leq 100$$

$$20x + 60y \leq 3000 \quad \text{OE}$$

B1

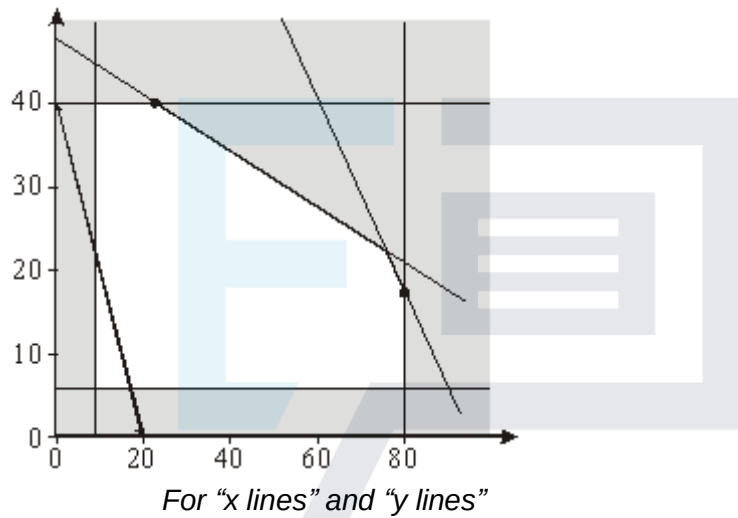
$$(\text{maximise})(P \Rightarrow) 2x + y$$

May be seen in (b) or (c)

B1

5

(b)



B1

{ For each other line M1 - ve gradient (0,50)
M1 - ve gradient (100,0)

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M1A1

M1A1

Feasible region correct to within 1 square

B1

Objective line

B1

7

(c) Max at (80,20)

Considering an extreme point in their region

M1

$$P = £180$$

region

A1

2

- (d) $P = x + 4y$
Max at (30, 40)

M1

$$P = £190$$

Using (30,40) ($\pm$ square)

A1

2

[16]

Q19.

- (a) $4x + 2y \leq 5 \times 4 \times 60$

Condone =

B1

1

- (b) $x \geq 40, y \geq 40$

Both

B1

$$x + y \geq 120$$

$$x + y \leq 400$$

Both

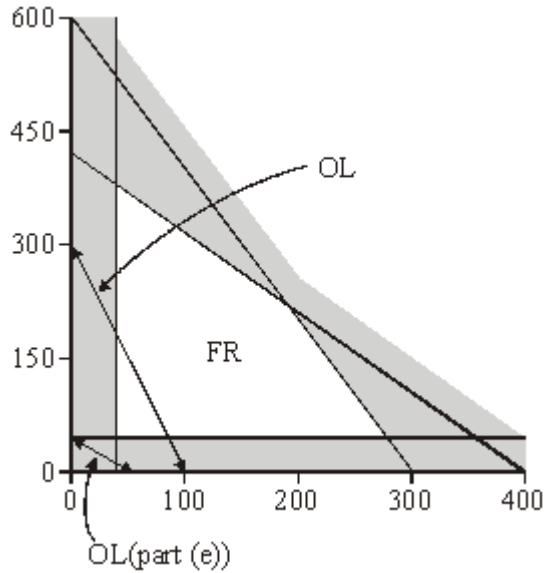
B1

(P =) $3x + y$ Exam Papers Practice. All rights reserved.

B1

3

- (c)



$$x \geq 40, y \geq 40$$

$$120 \leq x + y \leq 400$$

$$2x + y \leq 60$$

Correct FR

Correct OL

B1

B1

B1

B1

B1

5

(d) Extreme points

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M1

Max at $x = 280, y = 40$

A1

$$P = 840 + 40 = \text{£}880$$

SC: (280, 20) scores 1/3

B1

3

(e) (i) Max at (200, 200) $\rightarrow$ (40, 360)

M1

Profit £800

A1

2

(ii) No of combinations

B1

$$200 - 40 = 160$$

$$\begin{array}{r} + 1 \\ \hline 161 \end{array}$$

B1

2

[16]

Q20.

- (a) Milky
 $12x + 18y \leq 600$
 $\Rightarrow 2x + 3y \leq 100$

B1

1

- (b) $x \geq 15, y \geq 15$

B1

$$x + y \geq 35$$

B1

$$20x + 10y \leq 600$$

OE

B1

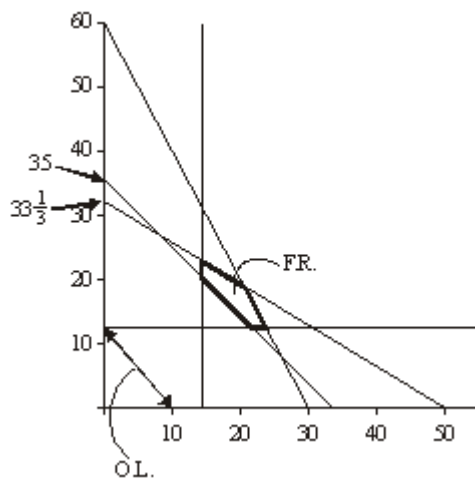
$$(P =) 1.5x + y$$

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B1

4

(c)



$$x \geq 15, y \geq 15$$

B1

Other 3 lines

B1 × 3

Feasible region

B1

Objective Line

B1

6

- (d) Considering one of their extreme points

M1

$$P = 50$$

A1

2

[13]

Q21.

(a)

$$\left. \begin{array}{l} \text{At least 30 \& 60} \\ \text{In total } \leq 200 \end{array} \right\}$$

B1

$$\text{Area } 2x + 3y \geq 300$$

B1

$$\left. \begin{array}{l} \text{Cost } 40x + 12y \leq 3600 \\ 10x + 3y \leq 900 \end{array} \right\}$$

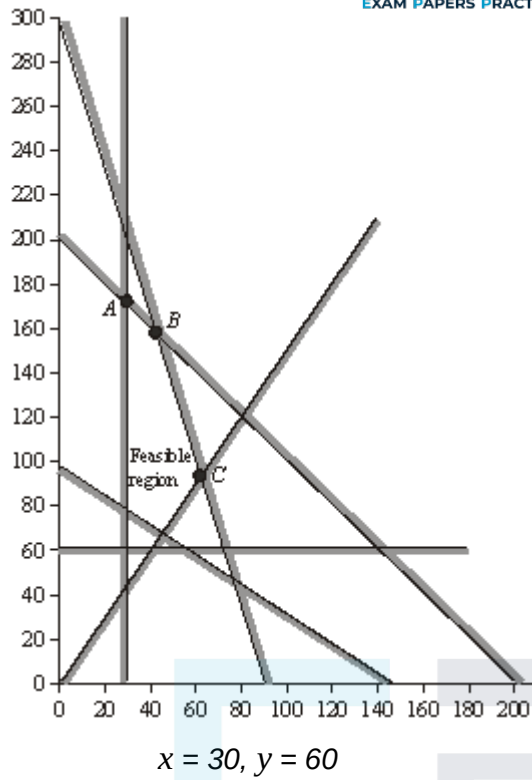
B1

$$\begin{array}{l} y \text{ at least } 150\% x \\ \text{(strict)} \end{array}$$

B1

4

(b)



other lines

Region

B1

B1×4

B1

6

(c) $y \geq 60$

B1

1

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(d) $P = 4x + 5y$
Max at A

Considering extreme point(s)

M1

$$\left. \begin{array}{l} x = 30 \\ y = 170 \end{array} \right\}$$

A1

$$P = 970$$

B1

3

[14]

Q22.

(a) $y \geq \frac{1}{2}x \Rightarrow 2y \geq x$

B1

$20x + 10y \leq 600 \quad \text{so} \quad 2x + y \leq 60$

B1

2

(b) $x \geq 5, y \geq 5, y \leq 25$

all correct

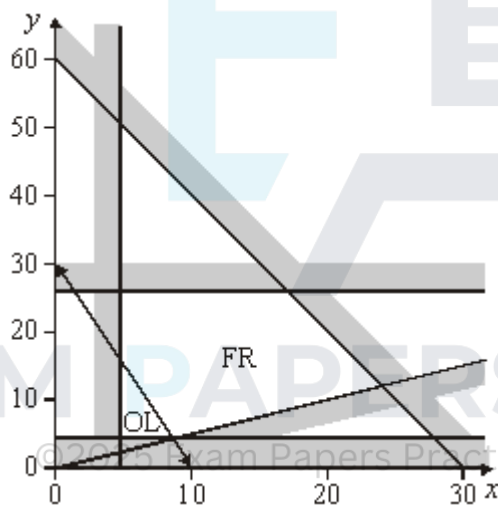
B1

$P = 3x + y$

B1

2

(c)



diagonal lines

B1 × 2

$x \geq 5, y \geq 5, y \leq 25$

B1

feasible region

B1

objective line

B1

5

(d) Max at $x = 24, y = 12$

M1

$$P = 84$$

A1

2

- (e) $P = 2x + 2y$
Max at (17.5, 25)

M1

Not integers
Values (17, 25) and (18, 24)

B1,B1

give $P = 510$ (6 hrs) + 84 (1hr)
= £594

CAO

A1

4

[15]

Q23.

- (a) $x \geq 2, y \geq 5, z \geq 2$

All

B1

$$x + y + z \leq 28$$

B1

$$40 \leq 3x + 2y + 2z \leq 60$$

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B1+1

4

- (b) (i) $x = z$

$$\therefore x \geq 2, y \geq 5$$

$$2x + y \leq 28$$

B1(*)

$$40 \leq 5x + 2y \leq 60$$

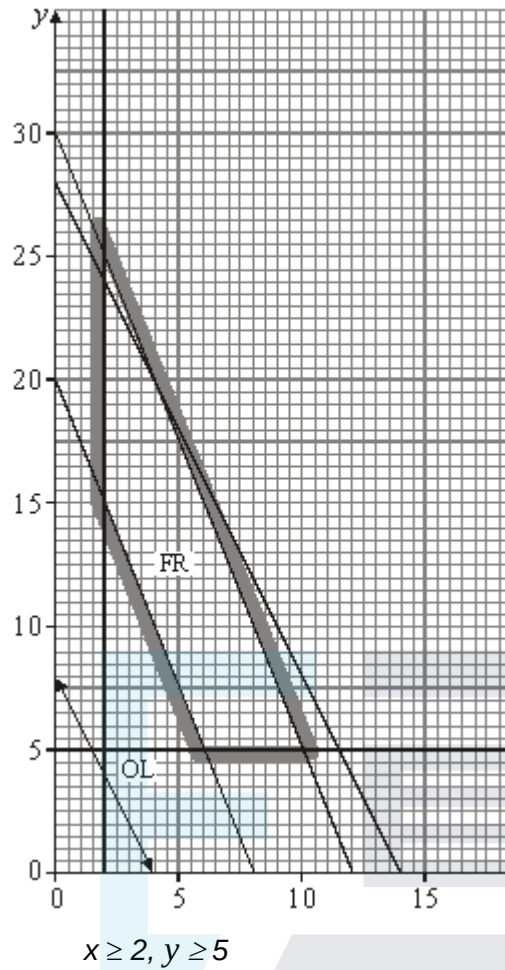
(*) Must have shown

$$x + z = 2x \text{ etc}$$

B1(*)

2

- (ii)



other lines

B1

region (pentagon)

B1 $\times$ 3

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objective line

M1

Correct

A1

7

(iii) Iron 28

$$\therefore x = 2, y = 24$$

Attempt at a max extreme point on their graph

M1

$$\text{or } x = 3, y = 22$$

$$\text{or } x = 4, y = 20$$

any pair

or 2,2,24

3,3,22
4,4,20

A1

2

[15]

Q24.

(a) $3x + 6y \leq 600$
 $\Rightarrow x + 2y \leq 200$

or roses

$6x + 3y \leq 600$
 $\Rightarrow 2x + y \leq 200$

or carnations

$4x + 4y \leq 480$

or lilies

$\Rightarrow x + y \leq 120$

OE

$x \geq 0, y \geq 0$

(condone absence)

B1 × 3

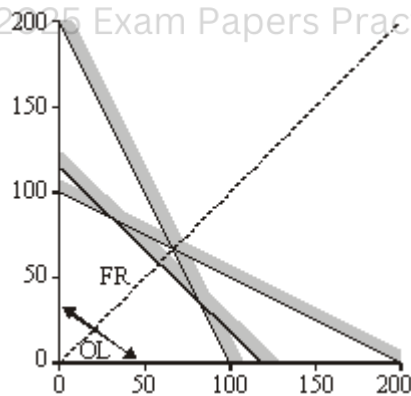
3

(b) $P = 1.5x + 2.5y$

(maybe shown in part (c))

B1

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Lines

B1 × 3

Region

B1

Objective line with -ve gradient

M1

A1

7

(c) Checking an extreme point

M1

$$P = 260$$

A1

2

(d) $x > y$

or $y = x$ on diagram

M1

Max at (60, 60)

Attempt at points around (60, 60)

M1

But $x > y$

$$\therefore x = 61, y = 59$$

A1

$$P = 239$$

A1

4

EXAM PAPERS PRACTICE [16]

Q25. ©2025 Exam Papers Practice. All rights reserved.

(a) $x \geq 7, y \geq 7$

Both

B1

$$y \leq 2x$$

B1

$$y \geq \frac{1}{3}x$$

B1

$$2y + x \leq 40$$

OE; M1 for – ve gradient

M1A1

$$3x + 4y \geq 60$$

OE; M1 for – ve gradient

M1A1

7

(b) Max at (8, 16)

M1A1

$$= 96$$

M1 Considering extremes

A1

3

[10]

Q26.

$$x \geq 10$$

$$y \geq 10$$

$$z \geq 10$$

all three

B1

$$x + y + z \geq 100$$

B1

$$2x + 5y + 3z \leq 400$$

B1

$$y \geq \frac{2}{5}(x + y + z)$$

M1

$$\Rightarrow 3y \geq 2$$